

EnergyDNA: A Blockchain-Based Traceability Framework for Secure and Transparent Renewable Energy Certification

Manichandan V

Department of Computer Science and Engineering
Ramaiah Institute of Technology
Bangalore, India

K. Akshay

Department of Computer Science and Engineering
Ramaiah Institute of Technology
Bangalore, India

Abstract—The rapid adoption of renewable energy sources has intensified the need for transparent, verifiable, and secure energy certification mechanisms. Existing Renewable Energy Certificate (REC) systems suffer from limitations such as lack of traceability, double counting, and vulnerability to fraud. This paper presents EnergyDNA, a blockchain-based framework that assigns a unique cryptographic identity to each unit of renewable energy using Non-Fungible Tokens (NFTs). The system leverages Ethereum smart contracts to ensure tamper-proof tracking of energy generation, transfer, and consumption. Additionally, an intelligent Energy Matchmaker algorithm is introduced to optimize energy allocation. Experimental results demonstrate low latency, high matching accuracy, and strong consistency between blockchain and database layers. The proposed system provides a scalable and trustworthy solution for next-generation energy certification.

Index Terms—Blockchain, Renewable Energy, NFTs, Ethereum, Smart Contracts, Traceability

I. INTRODUCTION

The transition toward renewable energy sources such as wind and solar plays a crucial role in achieving sustainable development and reducing environmental impact. However, one of the key challenges in this transition is ensuring that the energy being consumed is genuinely derived from renewable sources. Current certification mechanisms are largely centralized and lack transparency, making it difficult to verify the true origin of energy.

Traditional Renewable Energy Certificate (REC) systems attempt to address this issue, but they suffer from several limitations. These systems often operate on aggregated data, which makes it hard to trace energy back to its exact source. Additionally, they are vulnerable to problems such as double counting, delayed verification, and potential manipulation, which ultimately reduces trust among consumers and stakeholders.

Blockchain technology offers a promising solution to these challenges by introducing decentralization, immutability, and transparency. By maintaining a distributed and tamper-proof ledger, blockchain ensures that once data is recorded, it cannot be altered or misused.

Building on these advantages, EnergyDNA proposes a novel approach in which each unit of energy is assigned a unique

digital identity. This identity captures critical metadata such as generation time, source, and environmental conditions. As a result, it becomes possible to track energy from its point of generation to final consumption, ensuring complete lifecycle transparency and significantly improving trust in renewable energy systems.

II. RELATED WORK

Previous studies have explored the integration of blockchain technology in energy systems to improve transparency, security, and trust. Blockchain enables decentralized record-keeping through immutable ledgers and smart contracts, allowing automated and tamper-proof transactions. Several solutions have been proposed for renewable energy tracking and certificate management; however, most of them operate at an aggregated level, where energy is tracked in bulk rather than at the level of individual units.

This aggregation limits traceability and makes it difficult to verify the exact origin of consumed energy. EnergyDNA addresses this limitation by introducing a unit-level tokenization approach, where each energy unit is assigned a unique digital identity. By capturing detailed metadata and enabling lifecycle tracking from generation to consumption, the system improves granularity, enhances transparency, and builds greater trust in renewable energy ecosystems.

III. PROBLEM STATEMENT

Traditional renewable energy certification systems face several critical challenges that limit their effectiveness. One of the primary issues is the lack of transparency, as consumers and stakeholders often cannot verify the true origin of the energy they consume. Additionally, these systems are prone to double counting of certificates, where the same unit of energy may be claimed multiple times, leading to inaccuracies and potential fraud. Another significant limitation is the absence of end-to-end traceability, making it difficult to track energy from its point of generation to final consumption.

Furthermore, the reliance on centralized verification authorities introduces risks such as data manipulation, delayed

validation, and single points of failure. These limitations collectively reduce trust among users and hinder the scalability of renewable energy adoption, especially in large and distributed energy markets.

IV. PROPOSED SYSTEM

Each energy unit is converted into an NFT containing:

- **Turbine ID:** Identifies the specific wind turbine that generated the energy, ensuring source-level traceability.
- **Timestamp:** Records the exact time of energy generation, enabling chronological tracking and verification.
- **Wind Speed and Direction:** Captures environmental conditions during generation, providing contextual information about energy production.
- **Energy Output:** Represents the amount of energy generated (in kW/kWh), forming the core value of the token.
- **EnergyDNA Hash:** A unique cryptographic fingerprint generated using input parameters, ensuring immutability and preventing duplication.

$$H = SHA256(ID + t + v + P) \quad (1)$$

V. SYSTEM WORKFLOW

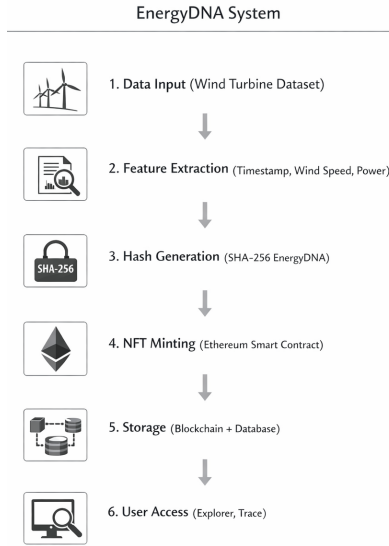


Fig. 1. EnergyDNA System Architecture

The system operates in four stages:

- **Data Acquisition from turbine dataset-** Collects wind turbine data such as timestamp, wind speed, and power output.
- **Hash Generation for EnergyDNA-** Generates a unique cryptographic hash for each energy unit using its metadata.
- **NFT Minting on blockchain-** Converts each energy unit into an NFT and records it on the blockchain.
- **Token Lifecycle Tracking-** Monitors the token through its stages of minting, transfer, and retirement.

VI. SYSTEM ARCHITECTURE

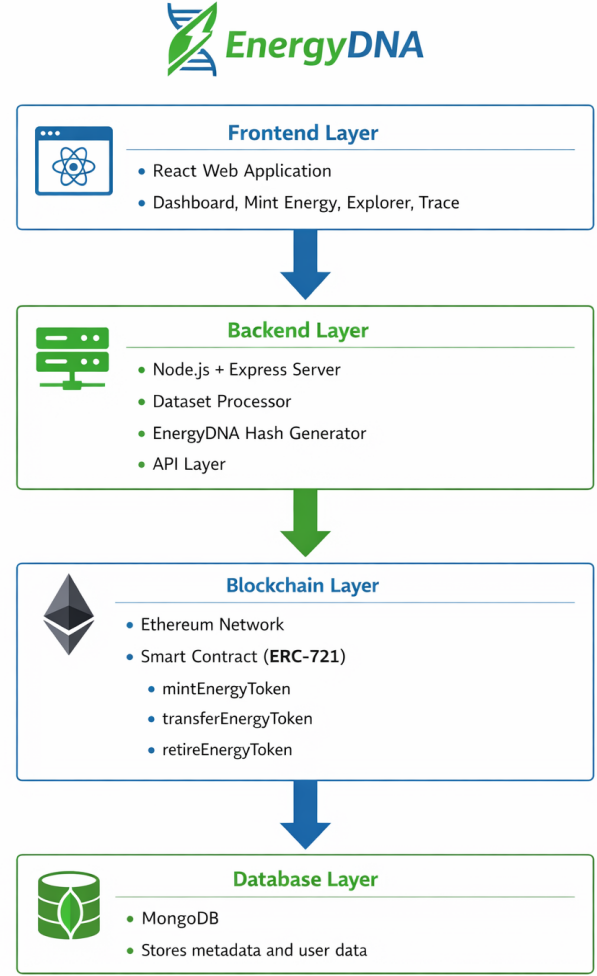


Fig. 2. EnergyDNA System Architecture

The system consists of frontend (React), backend (Node.js), blockchain (Ethereum), and database (MongoDB).

VII. ENERGY TOKEN LIFECYCLE

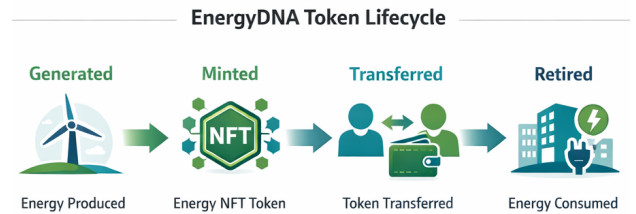


Fig. 3. Energy Token Lifecycle

- Generated
- Minted
- Transferred
- Retired

VIII. IMPLEMENTATION DETAILS

The EnergyDNA system is implemented using a full-stack architecture that ensures efficient data processing and user interaction. The frontend is developed using React.js with Tailwind CSS to provide a responsive and intuitive user interface. The backend is built using Node.js and Express.js, which handle API requests, dataset parsing, and communication between system components.

Blockchain integration is achieved through Ethereum smart contracts deployed on a test network, following the ERC-721 standard for representing energy units as non-fungible tokens. To efficiently process large datasets (over 50,000 records), a stream-based parsing approach is used, minimizing memory usage and improving performance. Each data entry is transformed into a unique energy token, enabling secure and traceable representation of energy units within the system.

IX. BLOCKCHAIN INTERACTION MODEL

The interaction between the application and the blockchain is facilitated using Web3.js, which enables seamless communication between the frontend and the Ethereum network. When a user initiates the minting process, the frontend generates a transaction request that is signed and submitted through MetaMask, ensuring secure user authentication and transaction authorization.

Once the transaction is received, the smart contract validates the request and records the corresponding energy token along with its associated metadata on the blockchain. Each transaction produces a unique hash, which serves as a verifiable record, ensuring both traceability and immutability of the data.

Ownership transfer of tokens is managed through secure smart contract function calls, allowing controlled and transparent exchange between users. Additionally, the retirement mechanism ensures that once an energy unit is consumed, its corresponding token is permanently marked as inactive, preventing reuse and maintaining system integrity.

X. CONSISTENCY MODEL

EnergyDNA maintains consistency between blockchain and database layers through a dual-write mechanism. Every successful blockchain transaction is mirrored in the database.

In case of transaction failure, rollback mechanisms ensure system integrity. This guarantees that both layers remain synchronized, preventing data inconsistency.

XI. ENERGY MATCHMAKER ALGORITHM

```

Input: Required Energy R
Search dataset for exact match
if exact match exists then
    return match
else
    find closest or combination
end if

```

XII. ENERGY TOKEN GENERATION ALGORITHM

```

Input: Turbine Dataset D
for each entry e in D do
    Extract turbine ID, timestamp, wind speed, power
    Generate EnergyDNA:
     $H = SHA256(ID + t + v + P)$ 
    Create metadata object M
    Call smart contract:
    mintEnergyToken(M)
    Store token ID and metadata
end for
Output: Set of Energy Tokens

```

XIII. MATHEMATICAL MODEL

$$E = f(v, t) \quad (2)$$

$$H = SHA256(ID + t + v + P) \quad (3)$$

XIV. PERFORMANCE EVALUATION

TABLE I
SYSTEM PERFORMANCE METRICS

Operation	Avg Latency	Peak Latency
Dataset Load	18.5 ms	29.4 ms
Matching	20.6 ms	24.4 ms
Minting	288.0 ms	449.6 ms
DB Sync	18.1 ms	30.3 ms

A. Performance Analysis

The performance metrics presented in Table I are obtained through real-time experimental evaluation of the EnergyDNA system. The dataset loading operation exhibits low latency (average 18.5 ms), demonstrating efficient stream-based parsing for large datasets.

The matching operation also shows low latency (20.6 ms on average), indicating that the Energy Matchmaker algorithm supports fast lookup and real-time response, making it suitable for dynamic energy allocation scenarios.

Blockchain minting has comparatively higher latency (288 ms on average) due to transaction processing and network confirmation overhead. However, the latency remains within acceptable limits for practical deployment.

Database synchronization is highly efficient, with an average latency of 18.1 ms, ensuring fast backend operations and strong consistency between blockchain and database layers.

Overall, the results indicate that EnergyDNA achieves a balance between performance, scalability, and security, making it suitable for real-world applications. The metrics were measured using real-time execution logs, with multiple runs averaged for accuracy, on a system with Node.js, MongoDB, and an Ethereum test network using MetaMask.

B. Practical Implications

The observed latency values indicate that EnergyDNA can operate in near real-time environments. The low response time of matching operations makes the system suitable for dynamic energy markets, where rapid decision-making is required.

Although blockchain operations introduce moderate latency, they provide strong guarantees of security and immutability, which are critical for certification systems. The trade-off between latency and security is acceptable in the context of renewable energy traceability.

These results demonstrate that EnergyDNA is not only theoretically sound but also practically deployable in real-world energy infrastructures.

XV. COMPARISON WITH TRADITIONAL REC

TABLE II
DETAILED COMPARISON WITH TRADITIONAL REC SYSTEMS

Feature	Traditional REC	EnergyDNA
Traceability	Limited (Batch-Level)	Full (Unit-Level)
Transparency	Low	High (Blockchain)
Fraud Risk	High (Manual)	Very Low (Immutable)
Granularity	Aggregate data	Individual units
Verification Time	Delayed	Near Real-Time
Data Integrity	Moderate	High (Cryptographic)
Ownership	Limited	Lifecycle tracking
Tracking		
Automation	Low	High (Smart Contracts)
Scalability	Moderate	High (Layer-2)

Table II highlights the key differences between traditional REC systems and the proposed EnergyDNA framework. Traditional systems operate on aggregated data, limiting traceability and often resulting in delayed verification processes.

In contrast, EnergyDNA provides unit-level tracking, enabling precise identification of energy origin. The use of blockchain ensures immutability and significantly reduces the risk of fraud.

Additionally, EnergyDNA supports near real-time verification and automated processes through smart contracts, improving efficiency and reducing manual intervention. These advantages make EnergyDNA a more robust and scalable solution for modern renewable energy ecosystems.

XVI. SECURITY ANALYSIS

EnergyDNA ensures tamper-proof record management by leveraging the immutability of blockchain technology, where once data is recorded, it cannot be altered or deleted. The use of NFTs guarantees uniqueness of each energy unit, effectively preventing duplication or double counting. Additionally, secure ownership tracking is maintained through blockchain-based transaction records, enabling transparent and verifiable transfer of energy tokens between users.

XVII. SCALABILITY CONSIDERATIONS

The scalability of EnergyDNA is influenced by factors such as blockchain transaction throughput and associated gas costs, which can impact system performance at larger scales. To mitigate these challenges, the system can integrate Layer-2 solutions such as Polygon or Optimistic Rollups, which provide faster transaction processing and reduced costs.

Additionally, techniques such as batch minting and off-chain storage of metadata can significantly improve scalability by reducing on-chain load. Future implementations may further enhance performance by adopting hybrid architectures that combine public and permissioned blockchains, balancing transparency with efficiency.

XVIII. REAL-WORLD APPLICABILITY

EnergyDNA can be effectively deployed in smart grids, renewable energy markets, and carbon credit systems, where transparency and accountability are critical. By enabling consumers to verify the origin and attributes of their energy consumption, the system enhances trust and supports informed decision-making in sustainable energy usage.

Furthermore, the integration of EnergyDNA with IoT-enabled turbines allows real-time data capture and automated token generation. This capability makes the system well-suited for large-scale deployments, enabling continuous monitoring, accurate tracking, and seamless operation across distributed energy infrastructures.

XIX. FUTURE WORK

- IoT integration- Enables real-time data collection from turbines for automated and accurate energy token generation.
- Layer-2 scaling- Improves transaction speed and reduces gas costs by leveraging scalable blockchain solutions.
- AI-based prediction- Utilizes machine learning models to forecast energy generation and optimize resource allocation.

XX. CONCLUSION

EnergyDNA provides a scalable, secure, and transparent framework for renewable energy certification by leveraging blockchain technology and NFTs. It enables precise tracking of energy units, ensures data integrity through immutability, and enhances trust among stakeholders through verifiable and tamper-proof records.

ACKNOWLEDGMENT

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